PRST Class (07/10/2020)

Eg $E(X) = 5,\ V(X) = 3$

$P(|X-5| < h) \geq 0.99$

$\Rightarrow 1 - P(|X-5| \geq h) \geq 0.99$

$\Rightarrow P(|X-5| \geq h) \leq 0.01$

$P(|X-5| \geq h) \leq \frac{3}{h^2}$

$0.01 = \frac{3}{h^2} \Rightarrow h^2 = \frac{3}{0.01}$

Eg. $X \quad f(x) = \lambda e^{-\lambda x},\ x>0;\ \lambda>0$

find an upper bound of $P(|X - E(X)| \geq b)$ using Chebyshev's inequality.

$E(X) = \frac{1}{\lambda} \quad V(X) = \frac{1}{\lambda^2}$

$E(X) = \int_0^\infty x\lambda e^{-\lambda x}\,dx = \frac{1}{\lambda}$

$V(X) = E(X^2) - [E(X)]^2$

$= \frac{1}{\lambda^2}$

$P(|X - \frac{1}{\lambda}| \geq b) \leq \frac{1}{\lambda^2 b^2}$

Convergence in probability

$X_1, X_2, X_3, \dots\dots$ sequence of rvs.

" of rvs

$\{X_n\}$:

X — a rv

$\lim_{x \to a} f(x) = c$

X — a $\quad |x \to a$

If $\forall\, \varepsilon > 0$, $\lim_{n \to \infty} P[|X_n - X| > \varepsilon] = 0$

or

$\lim_{n \to \infty} P(|X_n - X| < \varepsilon) = 1$

then $\{X_n\}$ is said to converge to X in probability.

$X_n \xrightarrow{P} X$

X_1 : ~~throw~~ outcome coming due to 1st throw of a die

X_2 : " " " 2nd " that "

X_3 : " " " 3rd ...

X_4 :

⋮

X_{100} :

⋮

X_n → outcome coming due to nth throw of the die. $n = 1, 2, \ldots$

$\{X_n\}$ — sequence of rv.

$\lim_{n \to \infty} P[|X_n - X| < \varepsilon] = 1 \iff X_n \xrightarrow{P} X$

$\Omega = \{\omega_1, \omega_2, \ldots\}$

$P(|X_n - X| < \varepsilon)$

$= P(\omega : |X_n(\omega) - X(\omega)| < \varepsilon) \to 1$ as $n \to \infty$

$\Omega \to \mathbb{R}$

$X_n(\omega)$

Convergence almost surely :

Convergence almost surely.

$X_n \xrightarrow{a.s.} X$

$\lim_{n\to\infty} X_n(\omega) = X(\omega)$

$\forall \omega \in N \subseteq \Omega$

$P(N^c) = 0$

$X_n(\omega)$

$f(x)$

$\mathbb{R} \to \mathbb{R}$

$x \to f(x)$

$\lim_{n\to\infty} X_n = X$

$\lim_{x\to 0} f(x) = c$

Weak Law of Large number.

$X_1, X_2, X_3 \ldots\ldots$ sequence of rvs are independent and identically distributed. (i.i.d.)

$E(X_n) = \mu \quad \forall n$

$$\left(\frac{1}{n}\sum_{i=1}^{n} X_i \xrightarrow{P} \mu\right)$$

X_n: ~~ee~~ denotes the number coming due to the n th throw of the die

$P(X_n = 1) = \frac{1}{6}$

$P(X_n = 2) = \frac{1}{6}$

$P(X_n = 3) = \frac{1}{6}$

$\vdots$

$P(X_n = 6) = \frac{1}{6}$

$P(X_1 = 1) = \frac{1}{6}$, $P(X_2 = 1) = \frac{1}{6}$

$P(X_1 = 2) = \frac{1}{6}$ $\quad P(X_2 = 2) = \frac{1}{6}$

$\vdots$

$P(X_1 = 6) = \frac{1}{6}$ $\quad P(X_2 = 6) = \frac{1}{6}$

$Y_1, \; Y_2 \quad \{0, 1\}$

$P(Y_1 = 0) = 0.2$ | $P(Y_2 = 0) = 0.2$

$P(Y_1 = 1) = 0.8$ | $P(Y_2 = 1) = 0.8$

Y_1 and Y_2 have identical distribution